SOP-HARDWARE KITTING

[CONFIDENTIAL]
[UTTHUNGA]
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Contents

1 Purpose 3
2 Scope 3
3 Abbreviations 3
4 Terms and Definitions 4
5 Responsibilities 4
6 RACI 5
7 SIPOC 6
8 Overall Process Flow Chart 7
9 Process Description 8
10 Information Security requirements 9
11 References 9
12 Annexures 9
13 Forms 10

CHANGE HISTORY

This section describes the details of changes that have resulted in the current document.

SL. No.	Document Version No.	Change Description	Reason For change	Date Prepared By	Date Reviewed By	Date Approved By
01	0.1	Draft	Implementation of the Integrated Management System	Siva Prasad 27-Nov-23		
02	Issue B Version 1.0	Review comments fixed			Anitha N 7-Dec-23	Santosh. V 9-Dec-23
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4	Issue C Version 1.0	Review of document	New version update of IMS	Anitha N 01-08-2025	Santosh V 08-Aug-25	Sanmangala MS 14-08-2025

1 Purpose

1.1 The purpose of Kitting is to streamline and optimize workflow by assembling a pre-defined set of components, tools, or materials required for a specific task or project. This process aims to enhance efficiency, reduce assembly time, and minimize errors by ensuring that all necessary items are readily available in an organized manner, kitting is commonly used of all prototype or pilot build (pre-production) activity to facilitate smoother operations and improve overall productivity.

2 Scope

2.1 The scope of kitting involves determining the specific boundaries and parameters like Components inclusion, Use Cases, Frequency, size and complexity, integration with other process, scalability to improve the quality of the assembled PCB within an organization.

3 Abbreviations

a. PM - Project Manager

b. SVN - Source Version Control

c. ICT - Information Communications Technology

d. SLA - Service Level Agreements

e. RFQ - Request for Quotation

f. GRN - Goods Receipt Note

g. IR – Inspection Report

h. QMS – Quality Management System

i. APL – Approved Supplier List

j. OEM – Original Equipment Manufacturer

k. NDA – Non-Disclosure Agreement

l. ISMS– Information Security Management System

m. MPN—Manufacture part number

4 Terms and Definitions

a. Kitting: The process of assembling and organizing a set of pre-defined components, tools, or materials required for a specific task or project.

b. Components: Individual items or parts that make up the contents of a kit. These can include physical objects, tools, or materials.

c. Assembly Instructions: Detailed guidelines outlining the steps to assemble a kit, ensuring consistency and accuracy in the kitting process.

d. Quality Control: Procedures and checks implemented to ensure that components meet specified standards and that the assembled kits are free from defects.

e. Inventory Management: The systematic control and tracking of components and finished kits to maintain accurate stock levels and prevent shortages or excesses.

f. Documentation: Records and paperwork associated with the kitting process, including assembly instructions, quality control forms, and any other relevant paperwork.

5 Responsibilities

a. Hardware Lead ensures the accuracy and quality of components by verifying that they match the specifications outlined in the bill of materials (BOM).

b. Below are the responsibilities from Lab Team:

- Properly assemble components according to the provided assembly instructions, maintaining consistency and following best practices.
- Conduct quality checks on assembled kits to identify and address any defects or discrepancies, ensuring that finished kits meet established standards.
- Maintain accurate records of inventory levels for both individual components and assembled kits, updating systems to reflect changes

in stock.

- Complete and maintain all necessary paperwork associated with the kitting process, including assembly records, quality control forms, and any other relevant documentation.
- Keep the kitting workspace organized and clean, optimizing efficiency and facilitating a smooth workflow.
- Ensure that all tools and equipment used in the kitting process are in good working condition, performing routine maintenance as needed.
- Adhere to safety protocols and guidelines, promoting a safe working environment for all personnel involved in the kitting process.
- Identify opportunities for process improvement and efficiency gains within the kitting workflow, proposing and implementing changes as appropriate.
- Stay informed about new components or changes in assembly procedures and provide training to team members when necessary.
- If applicable, manage batch kitting processes, coordinating the simultaneous assembly of multiple kits to optimize production efficiency.
- Address and document any defective or unusable components identified during the kitting process, following appropriate disposal or corrective action procedures.
- Generate and provide regular reports on kitting activities, including production volumes, quality metrics, and any issues encountered during the process.

6 RACI

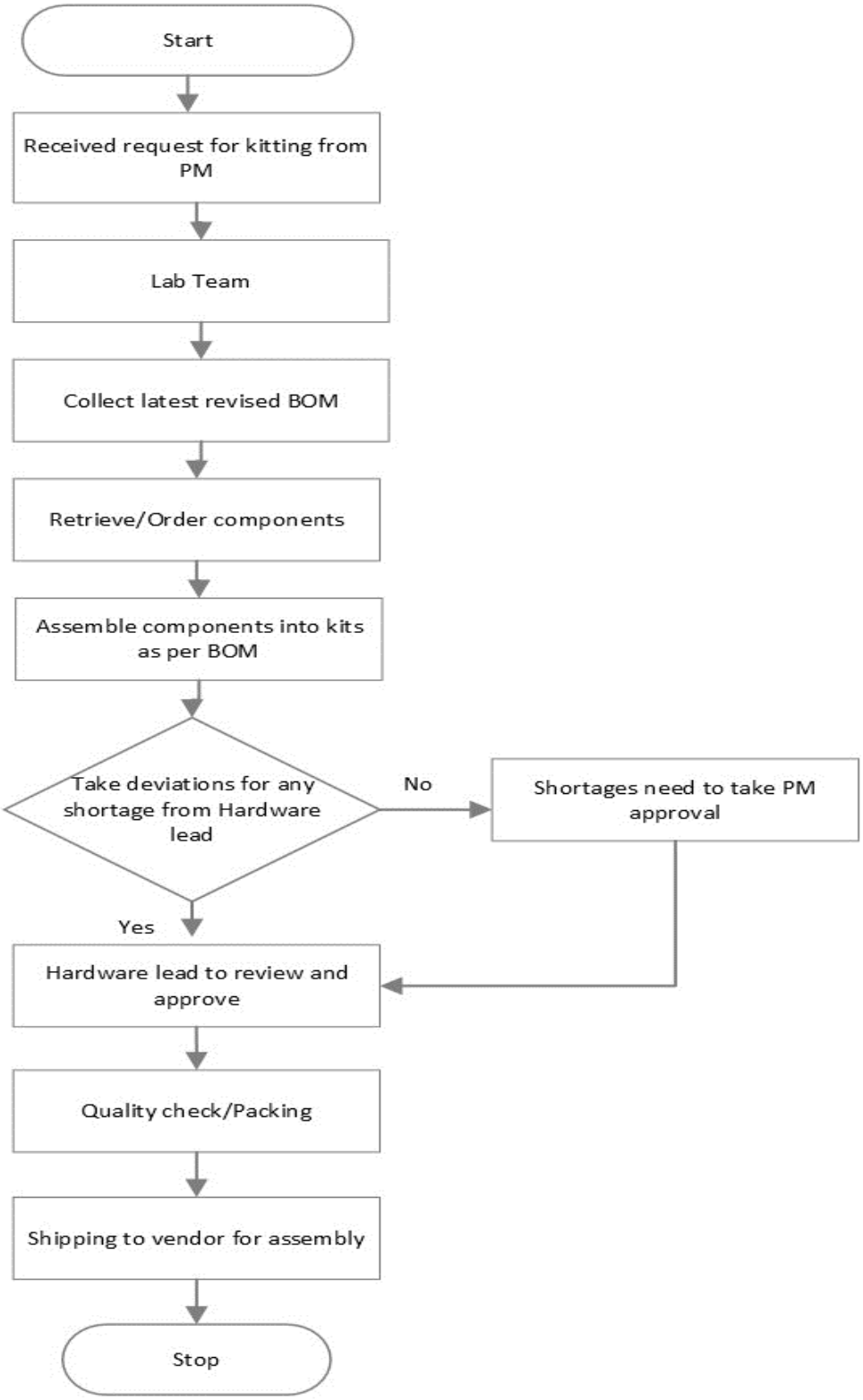
Process	Responsible	Accountable	Consulted	Informed
Define kit	Hardware team	Project Hardware Lead	Project Manager	Delivery Manager/Hardware Manager

Assemble kit	Lab Engineer	Lab manager	Hardware manager	Project manager
Verify kit	Hardware Engineer	Hardware lead	Hardware manager	Project manager
Deliver kit	Admin	Hardware Engineer	Project manager	Vendor

7 SIPOC

Supplier	Input	Process and Control		Output	Customer
		Process	Control		
Customer (Internal/External)	Order specifications	Kitting process	Receive and review order details	Order confirmation	Internal Teams/End-users
Internal Warehouse/External Supplier	Bill of materials, Inventory status	Kitting process	Identify and gather required components from storage	Gathered components	Assembly Team
Internal Warehouse	Gathered components	Kitting process	Assemble components into kits, perform quality checks	Completed kits, Quality control reports	Quality Control Team
Quality Verification	Quality Control Team	Kitting process	Completed kits, Quality control reports	Verify kits against quality standards	Verified kits, Quality verification reports Internal Teams/End-users
Internal Warehouse	Verified kits, Packaging materials	Kitting process	Pack kits securely for delivery	Packaged kits, Delivery documentation	Logistics Team/End-users

8 Overall Process Flow Chart



9 Process Description

a. Electronic component kitting is a process used in manufacturing that refers to the physical act of arranging the components as per the BOM that must be delivered to the assembly house or EMS.

- For all assemblies built in a Batch Build process, material is "Kitted" in the storeroom, transported to the workstation for assembly.
- After the completion of assembly, material is returned in storeroom.
- The Kitting process starts with a "Pick List" being generated by the excel system based on the schedule of jobs to build a product.
- Material shall be picked from inventory. lot numbers and serial numbers are used for tracking. material is bagged & tagged with labels printed specifically for the job.
- Labels are added to identify the material, job order number, job quantity, and reference designators.
- The material shall be staged in the assembly area as WIP (work-in-process) until it is ready to be assembled.

b. Below is the process to be followed by Lab team.

- The lab team receives the kitting request (project details, QTY, assembly house details) from the project manager or person designated by project manager.
- Kitting request is reviewed and approved by hardware manager.
- bill of materials (BOM) shall be reviewed to identify components required for the kit.
- Ensure all necessary components are available in inventory.
- Verify components against the BOM to ensure accuracy and quality.
- Lab team collects the latest BOM from the respective team and prepare a kitting check list.

- Lab team shall add BOM items to kitting template and verify the line items by Checking the components availability / component readiness. If components are inaccurate or defective, replace or rectify as needed.
- Lab engineer shall start the kitting as per the latest BOM and recheck the components (QTY, MPN). assembly instructions provided in the BOM, or associated documentation shall be followed.
- Deviation shall be taken for shortage of the components from Hardware. Any special instructions for assembly shall be mentioned in checklist/ email which will be shared to vendor.
- Hardware manager shall review the Kitting checklist for any deviations or defects and final checklist will be printed and shared to vendor for reference.
- Material shall be packed in a good condition and will be shipped to supplier or in-house assembly.

10 Information Security requirements

a. Supplier shall be assessed on basis of adherence to information security requirements specified by the Utthunga.

b. Supplier shall sign the NDA with the organization if required. NDA is Applicable for all Service based Suppliers. It is also applicable when confidential Data such as Specifications, Drawings, Designs, Patterns is to be shared to the suppliers (Subcontractors, Consultants & Testing)

c. ICT (Information Communications Technology) based Suppliers shall also be monitored on agreed SLA's (Service Level Agreements)

11 References

a. ISO 9001 Clauses 8.4.1, 8.4.2, 8.4.3

b. ISO 27001 Clauses 8.1, Controls A15

c. ISMS Control Policies Related to Supplier Management

12 Annexures

a. NIL

13 Forms

a. Hardware Kitting Checklist UT/SD/HP/TMP/02